Operating Systems

CS-DS-AIT203

Prepared by

Dr. Abeer Hassan El Bakly

Methodology of working within this semester

1- the activities of lectures are established as three online quizzes, which are online quizzes.

2- each student is responsible about his mobile, internet for quiz, sending the quiz within internet.

3- each student can take snapshot of quiz because the damage will be happened in internet to save the absence. Because the absence is included in the quiz.

4- if the student doesn't follow the instructions of quiz, his mark will be reduced with the half (writing his name in Arabic and till fourth name, his student code).

5- all quizzes and others will be in the question bank.

6- our credit hours are two and the maximum mark is 150, our marks will be classified as following:

30 (med term) –90 (practical + activities in lecture) – 60 (final exam)

Resources

1- A. Silberschatz, p. B. Galvin and G. Gagne. “operating system concepts”,10 edition, Wiely

2- S.Pal, “OPERATING SYSTEMS”, All India Council for Technical Education Nelson Mandela Marg, Vasant Kunj, New Delhi, 110070, 2023

WHAT IS AN OPERATING SYSTEM?

WHAT IS AN OPERATING SYSTEM?

An operating system is ……

Is it only for Computers?

How about these machines?

Car

Airplane

Printer

Washing Machine

Toaster

Compiler

Etc.

It is A program that acts as an intermediary between a user of a machine and the machine

WHAT IS AN OPERATING SYSTEM?

<u>An operating system</u> is <u>A program</u> that acts as an intermediary between a user of a computer and the computer hardware

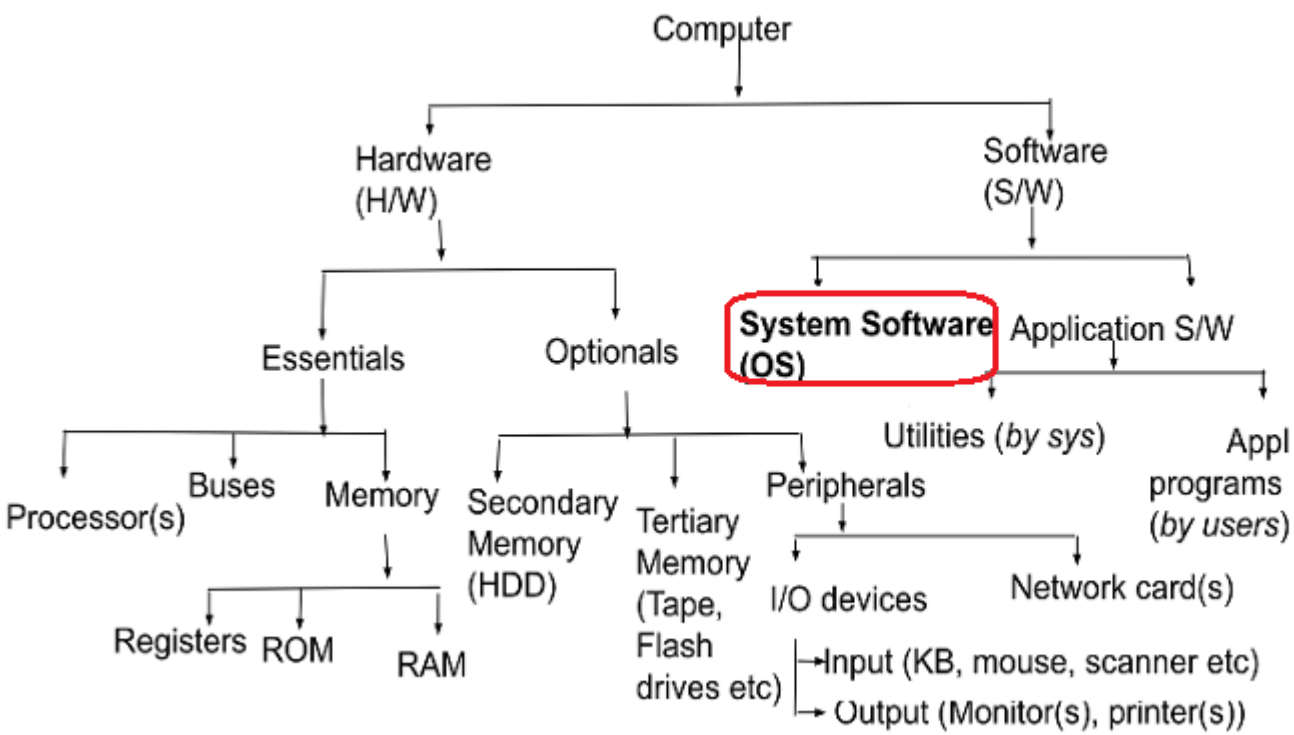


Fig. 1.2: Abstract view of a computer

WHAT IS AN OPERATING SYSTEM?

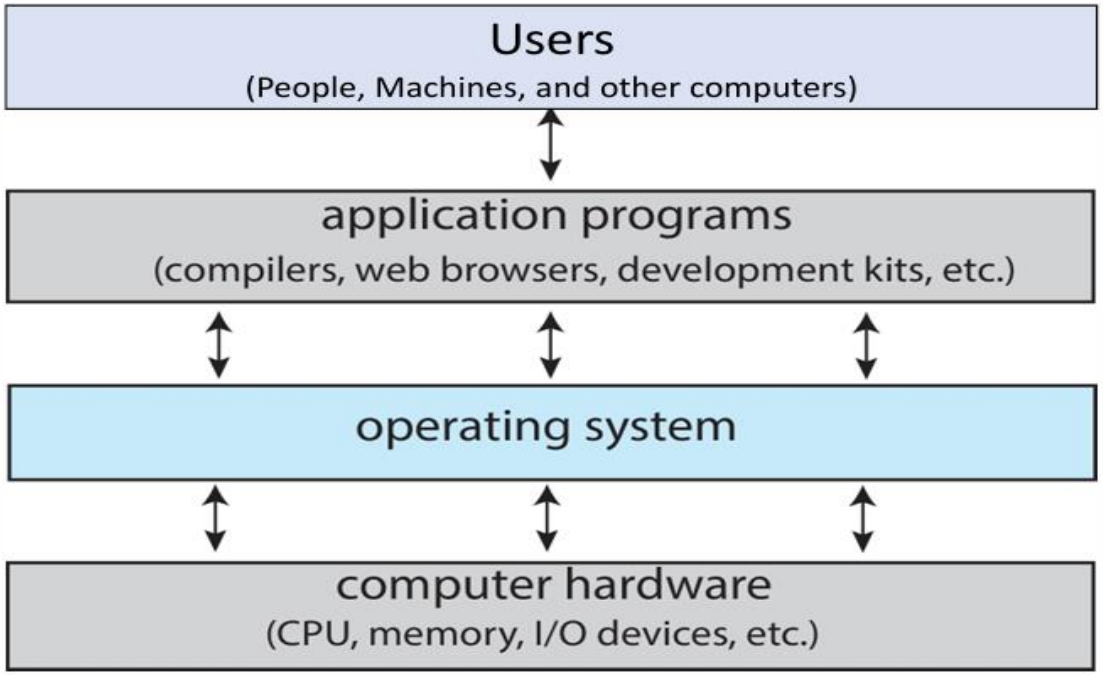
An operating System is a collection of system programs that together control the operations of a computer system.

Some examples of operating systems are UNIX, Mach, MS-DOS, MS-Windows, Windows/NT, Chicago, OS/2, MacOS, VMS, MVS, and VM.

Operating system goals:

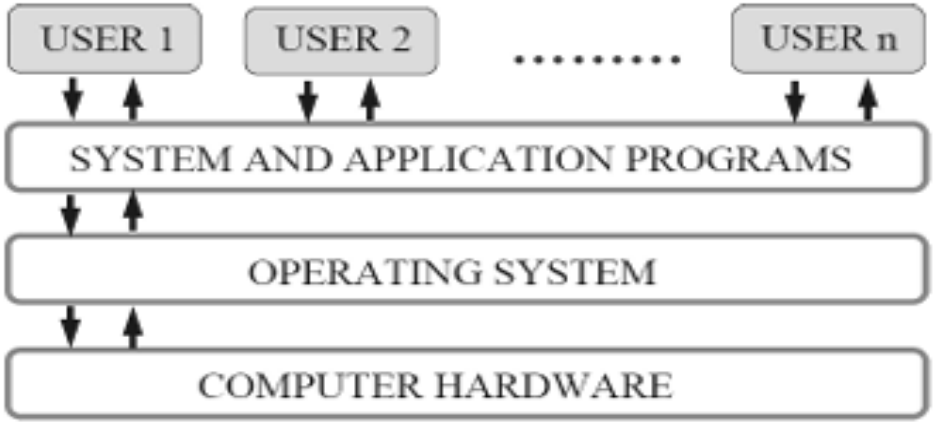
- **Execute user programs and make solving user problems easier.**
- **Make the computer system convenient to use.**
- **Use the computer hardware in an efficient manner.**

COMPUTER SYSTEM STRUCTURE



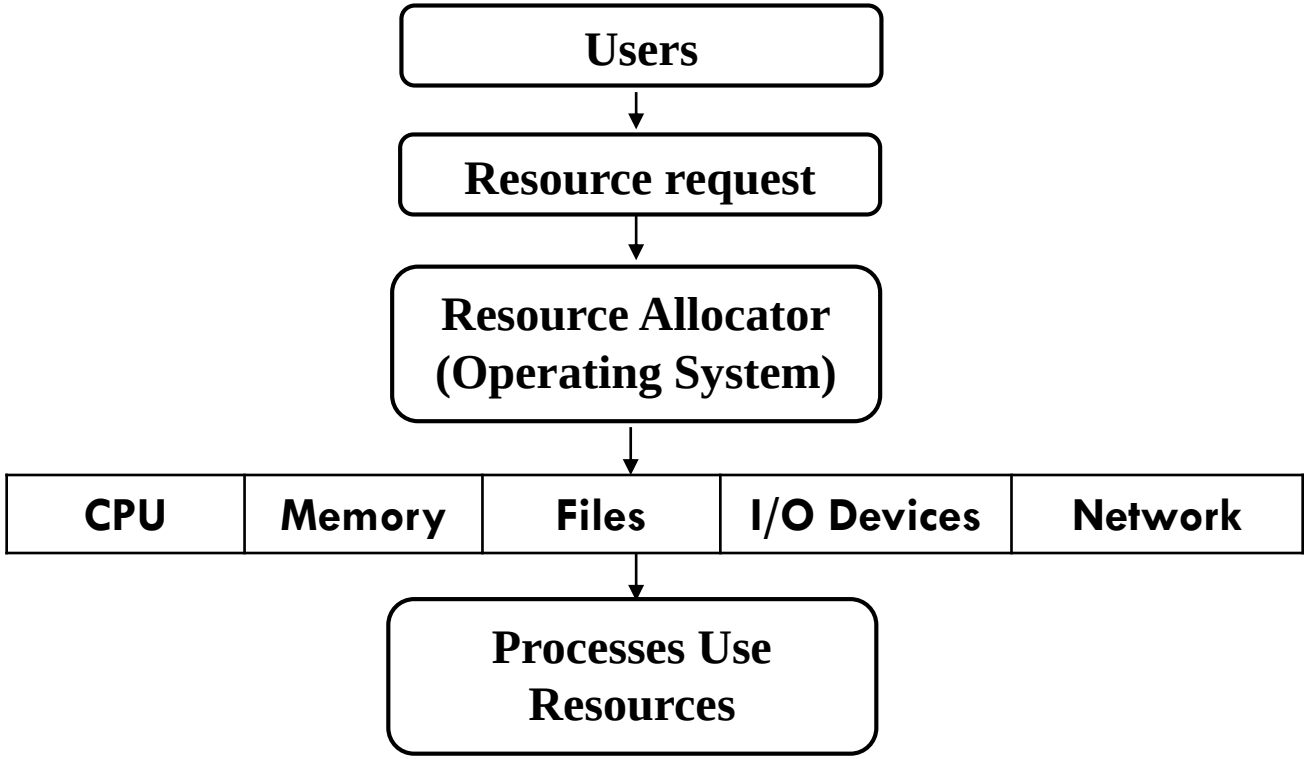
Computer System Components

1. **Hardware – provides basic computing resources (CPU, memory, I/O devices).**
2. **Operating system – controls and coordinates the use of the hardware among the various application programs for the various users.**
3. **Applications programs – Define the ways in which the system resources are used to solve the computing problems of the users (compilers, database systems, video games, business programs).**
4. **Users (people, machines, other computers). Abstract View of System Components**



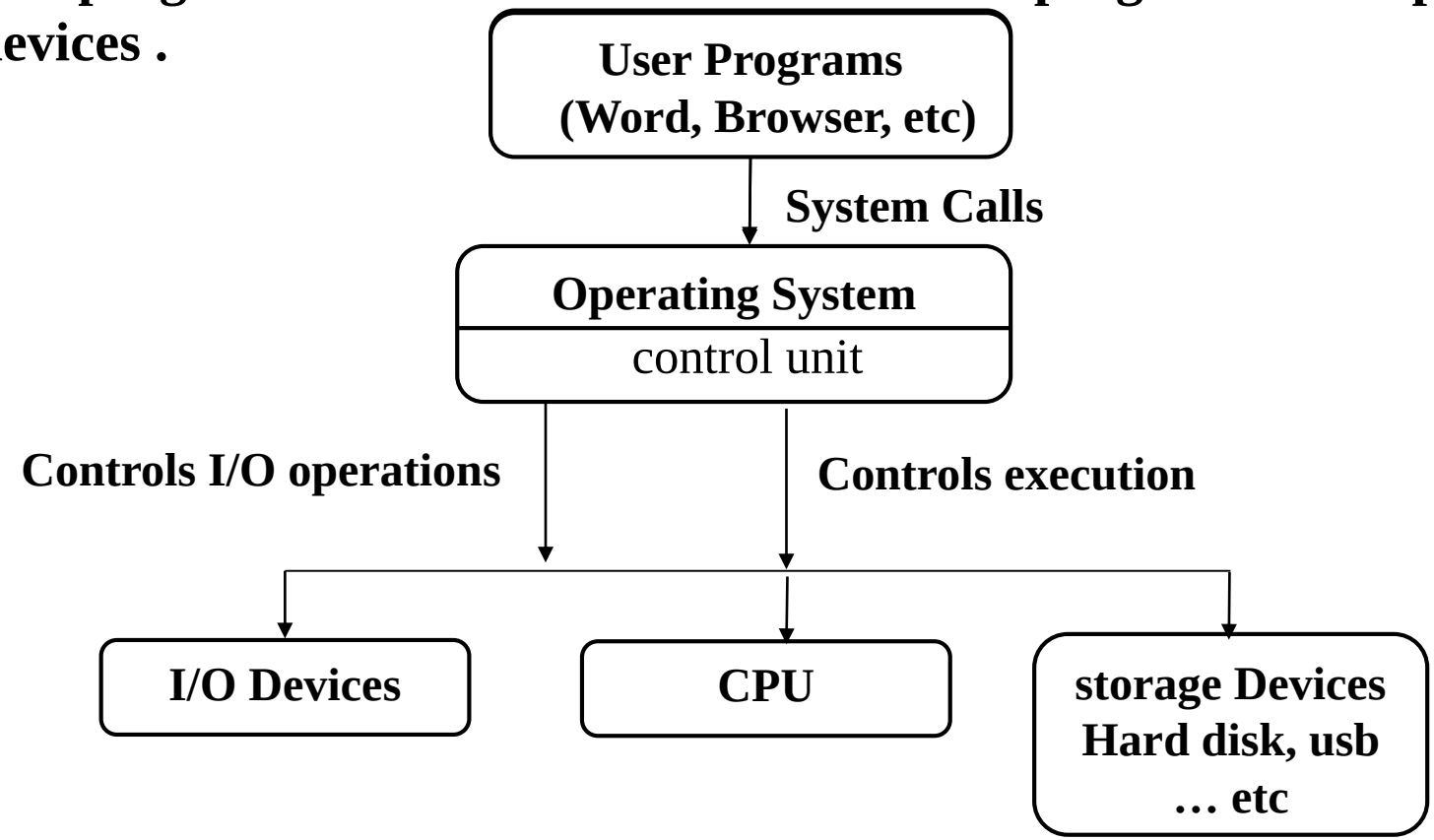
Operating System Definitions

Resource allocator : manages and allocates resources.



Operating System Definitions

Control program: controls the execution of user programs and operations of I/O devices .



Operating System Definitions

Control program: controls the execution of user programs and operations of I/O devices .

User Program Execution

- **User programs request services using system calls.**
- **OS loads programs into memory and schedules them on CPU.**
- **OS controls execution using process scheduling and context switching.**

Control of I/O Devices

- **OS manages all input/output operations.**
- **Uses device drivers to communicate with hardware.**
- **Allocates devices to programs and controls access.**

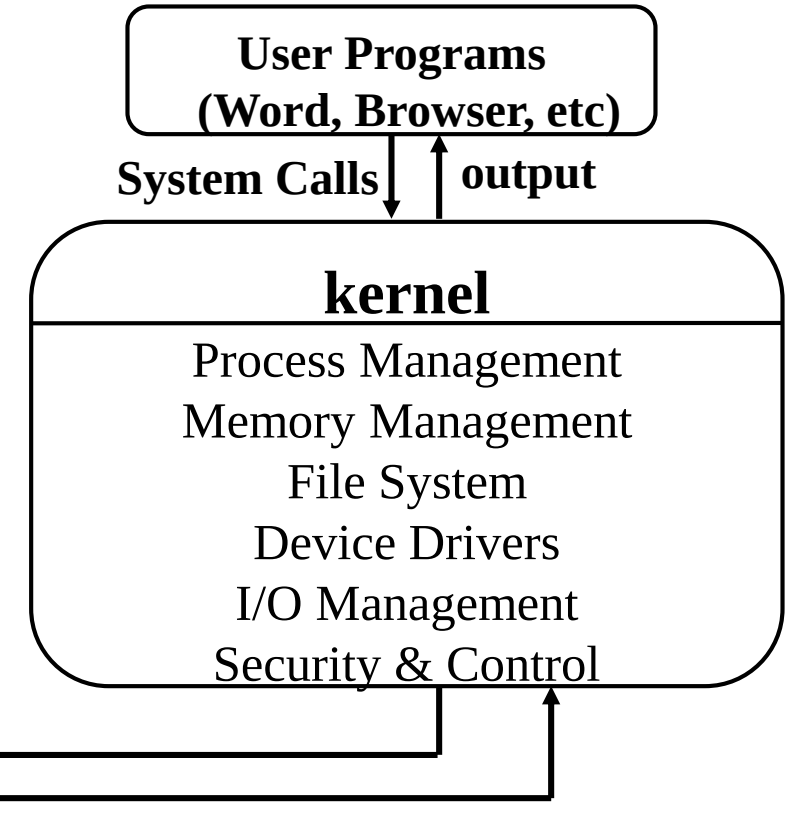
Role of OS

- **Acts as an interface between user programs and hardware.**
- **Ensures efficient and safe use of CPU and I/O devices.**
- **Prevents conflicts between multiple programs.**

Operating System Definitions

Components of OS: OS has two parts.

(1) Kernel is an active part of an OS i.e., it is the part of OS running at all times. It is a programs which can interact with the hardware. Ex: Device driver, dll files, system files etc.



- It acts as a bridge between applications (user level) and hardware.

It manages:

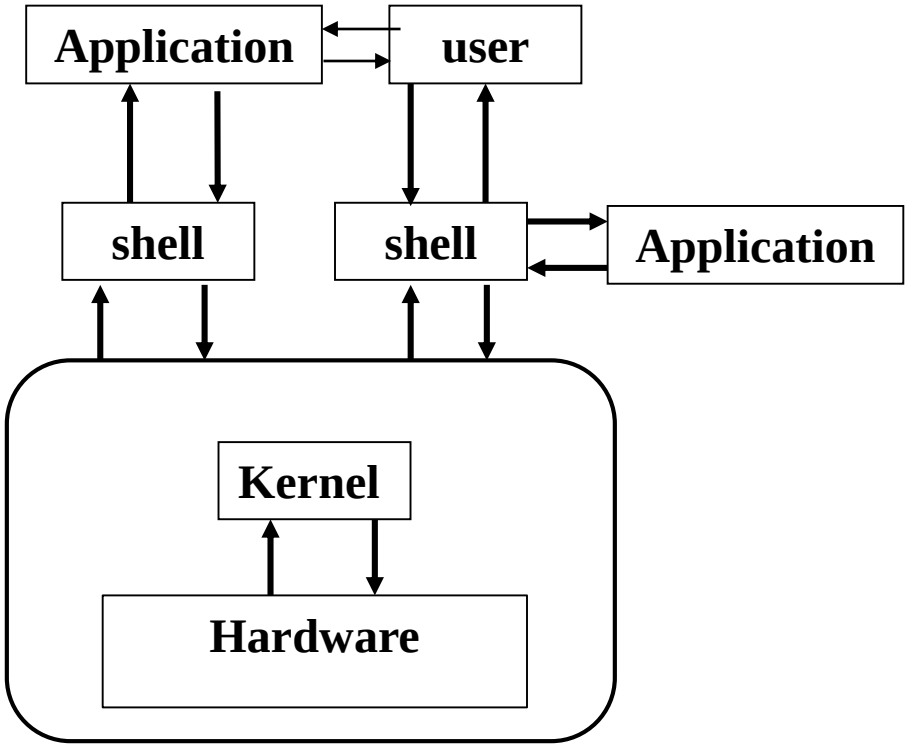
- Process management
- Memory management
- File system
- Device drivers
- Input/output operations

Operating System Definitions

Components of OS: OS has two parts.

(1) Shell is called as the command interpreter. It is a set of programs used to interact with the application programs. It is responsible for execution of instructions given to OS (called commands).

- The user enters commands.
- The shell interprets these commands.
- It sends instructions to the operating system.
- The OS executes the using application programs.
- The result is returned to the user.



Operating System Definitions

Operating systems can be explored from two viewpoints: the user and the system.

User View: From the user's point view, the OS is designed for one user to exploit its resources, to maximize the work that the user is performing and for ease of use.

System View: From the computer's point of view, an operating system is a control program that manages the execution of user programs to prevent errors and incorrect use of the computer. It is concerned with the operation and control of I/O devices.

Eovulation and types of OS:

Evolution and types of OS:

Types of OS in two views

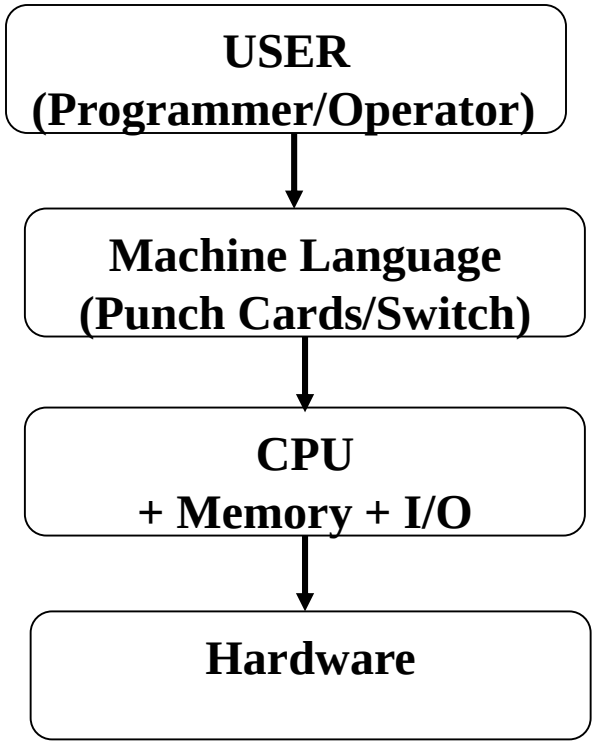
- Backend of OS
- Frontend of OS

Evolution and types of OS:

View for Backend of OS

1. 1940s–1950s: No Operating System

- Computers had no operating systems.
- Programs were executed manually using punch cards.
- Only one program ran at a time.



Evolution and types of OS:

View for Backend of OS

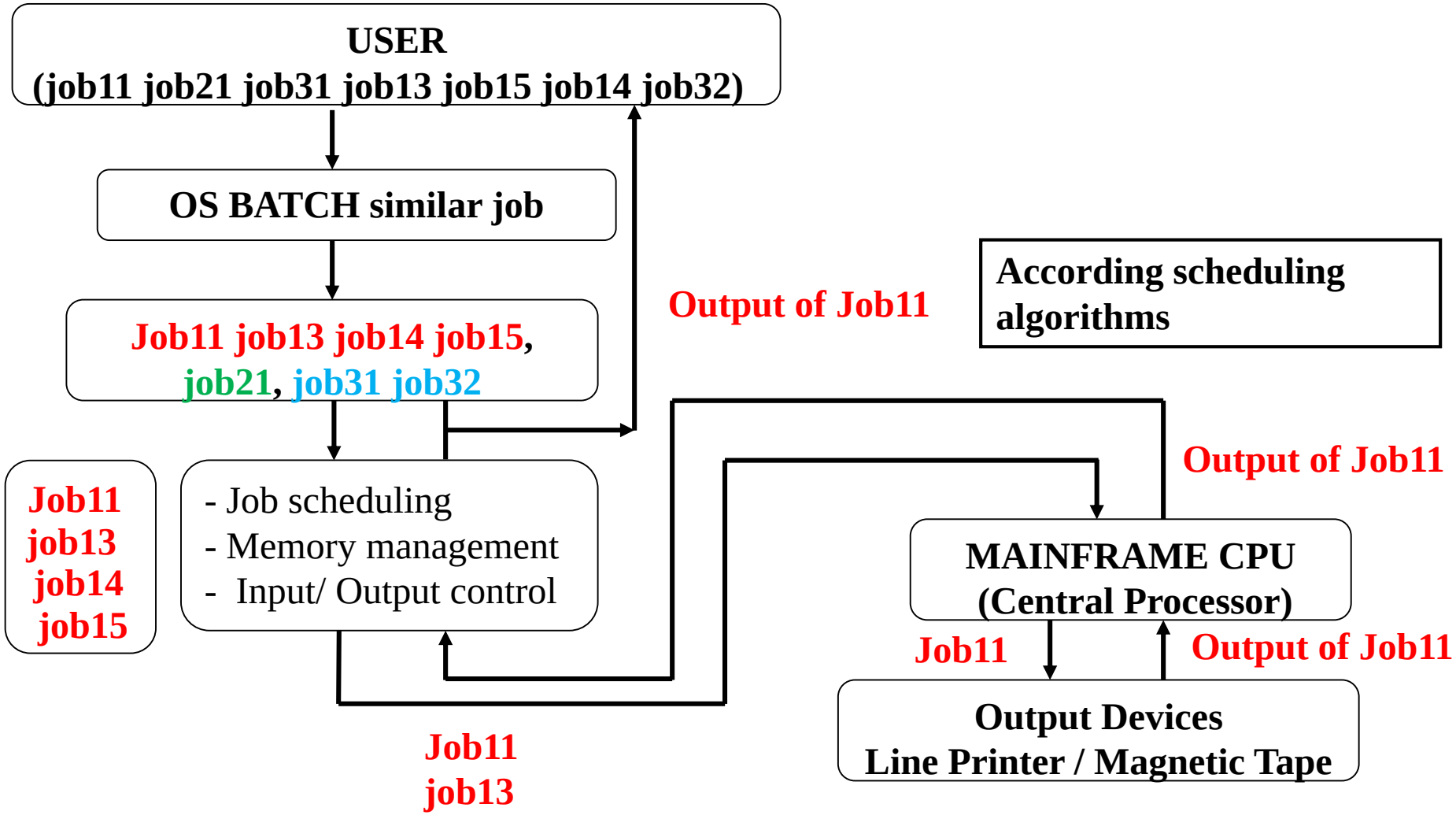
2. 1950s–1960s:Mainframe Systems

Reduce setup time by batching similar jobs Automatic job sequencing – automatically transfers control from one job to another.

- **First initial operating system. Resident monitor**
- **initial control in monitor**
- **control transfers to job**
- **when job completes control transfers pack to monitor**

Evolution and types of OS:

2. 1950s–1960s:Mainframe Systems



Evolution and types of OS:

View for Backend of OS

2. 1950s–1960s:Mainframe Systems

Advantages of Mainframe Operating System

- **Mainframe operating system have very high and massive processing power.**
- **They are scalable and can handle large scale of users and applications.**
- **They are fault tolerant and reliable.**
- **They provide great security measures to protect the system from hacking and other cyber security threats.**

Disadvantages of Mainframe Operating System

- **Mainframe operating system are very costly so only large institutions can afford them.**
- **They are very complex to operate. Specialized knowledge is required to use it.**
- **Mainframe Operating System aren't flexible.**

View for Backend of OS

3. 1960s–1970s Batch Processing Operating System:

- This type of OS accepts more than one jobs and these jobs are batched/ grouped together according to their similar requirements.

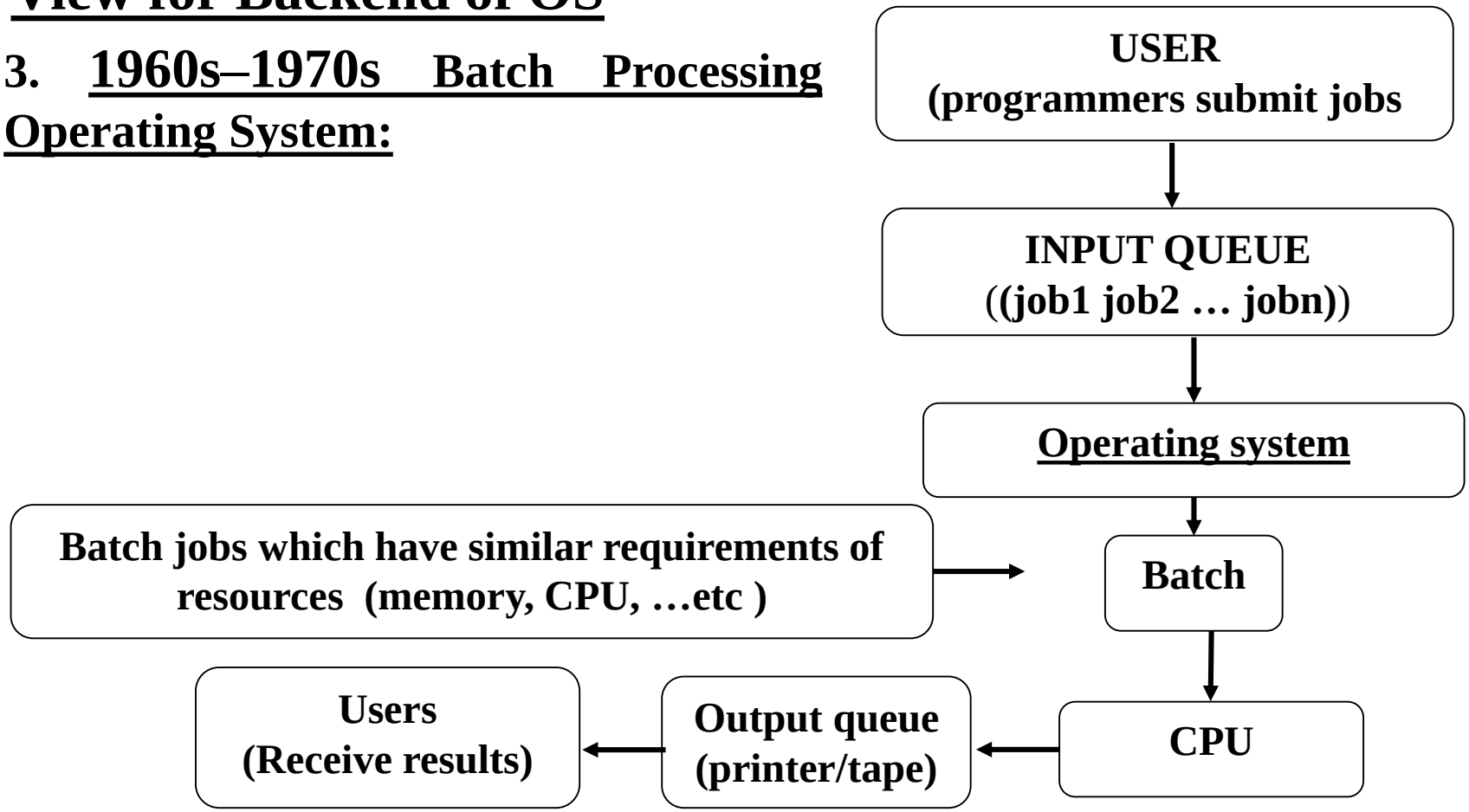
This is done by computer operator. Whenever the computer becomes available, the batched jobs are sent for execution and gradually the output is sent back to the user.

- It allowed only one program at a time.
- This OS is responsible for scheduling the jobs according to priority and the resource required.

Evolution and types of OS:

View for Backend of OS

3. 1960s–1970s Batch Processing Operating System:



View for Backend of OS

3. 1960s–1970s Batch Processing Operating System:

Advantages of Batch Operating System

Resource Efficiency: it improves the use of computation resources by processing jobs in groups and scheduling them during stages of resource accessibility.

High Throughput: it handles and complete a large number of tasks quickly, confirming quick turnaround times and high throughput.

Error Reduction: Since these systems work without requiring user interference, they minimize the risk of faults that can occur with manual processing.

Simplified Management: They restructure job management by automating the submission, scheduling, and implementation of tasks.

Cost Efficiency: By producing well-organized use of resources and reducing processing time and errors, batch processing systems can be a cost-effective option.

Scalability: it can manage a huge number of tasks, making them scalable and appropriate for large organizations with significant data processing needs.

View for Backend of OS

3. 1960s–1970s Batch Processing Operating System:

Disadvantages of Batch Operating System

Limited functionality: it can solve only simple tasks not solve more complex tasks. this can make them difficult to use for certain tasks, like managing files or software.

Security issues: they are not more secure because they are not typically used for day-to-day tasks, so they are not as secure as more common operating systems. This can lead to security risks if the system is used by people who should not have access to it.

Interruptions: it can be interrupted frequently, which can lead to missed deadlines or mistakes.

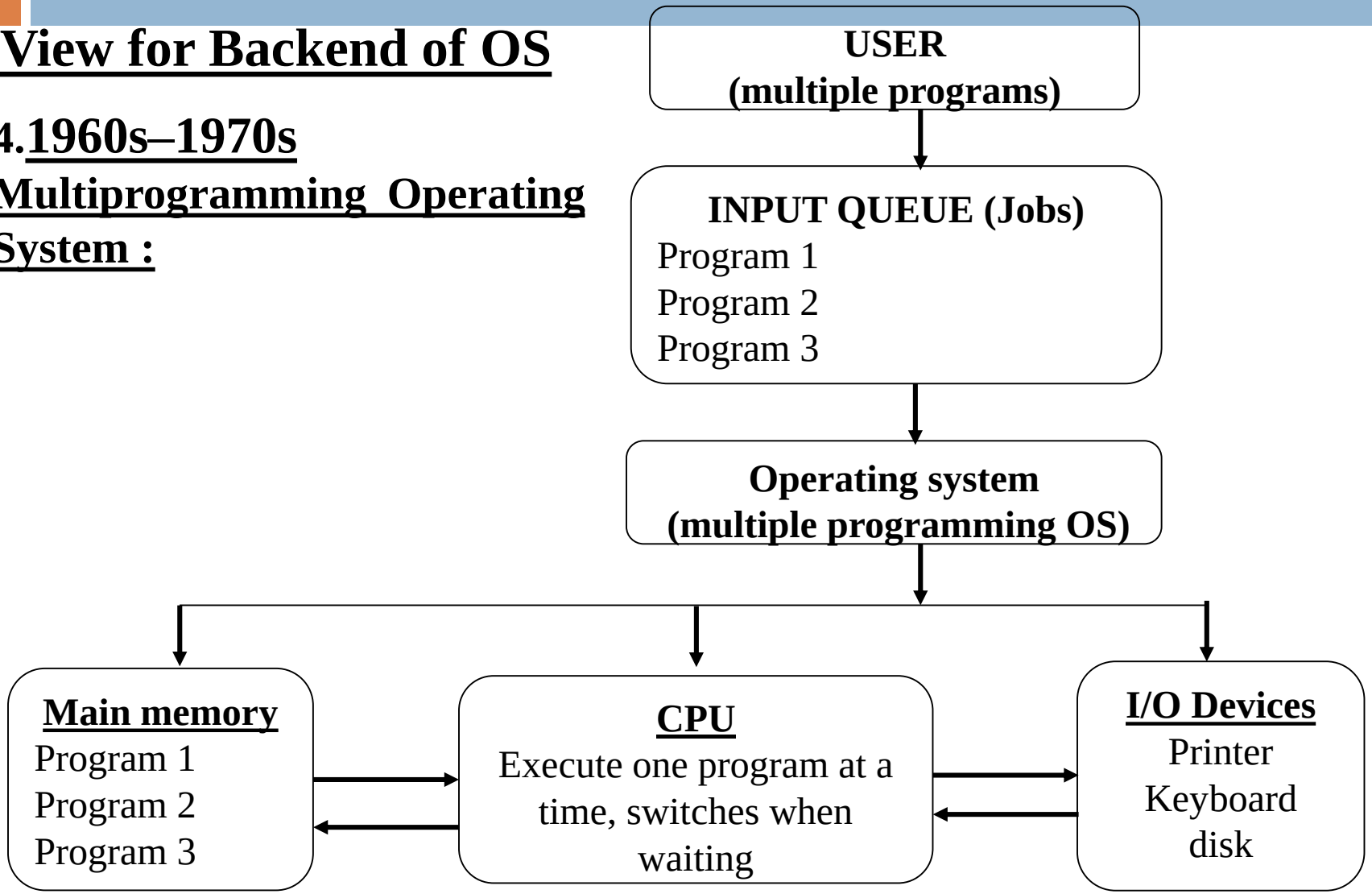
Inefficiency: they are often slow and difficult to use, which can lead to inefficiency in the workplace.

Evolution and types of OS:

View for Backend of OS

4.1960s–1970s

Multiprogramming Operating System :



Evolution and types of OS:

View for Backend of OS

4. 1960s–1970s Multiprogramming Operating System:

***The concept of multiprogramming is described as follows:**

- **All the jobs that enter the system are stored in the job pool(in disc). The operating system loads a set of jobs from job pool into main memory and begins to execute.**
- **During execution, the job may have to wait for some task, such as an I/O operation, to complete. In a multiprogramming system, the operating system simply switches to another job and executes. When that job needs to wait, the CPU is switched to another job, and so on.**
- **When the first job finishes waiting and it gets the CPU back.**
- **As long as at least one job needs to execute, the CPU is never idle. Multiprogramming operating systems use the mechanism of job scheduling and CPU scheduling.**
- **This type of OS is used to execute more than one jobs simultaneously by a single processor. it increases CPU utilization by organizing jobs so that the CPU always has one job to execute.**

View for Backend of OS

4. 1960s–1970s Multiprogramming Operating System:

Advantages of Multiprogramming

- Increase the resource utilization of a computer system.
- Support multiple simultaneously interactive users (terminals).
- Keeps multiple runnable jobs loaded in memory .
- Increase the throughput of the system.
- The waiting time for a program is reduced.
- Improve system's overall responsiveness.
- Improve the reliability and stability of the system.
- Make the system suitable for multitasking environments.

View for Backend of OS

4. 1960s–1970s Multiprogramming Operating System:

Disadvantages of Multiprogramming

- **If it has a large number of jobs, then long-term jobs will have to require a long wait.**
- **Memory management is needed in the operating system because all types of tasks are stored in the main memory.**
- **Using multiprogramming up to a larger extent can cause a heat-up issue.**
- **High Degree of Multiprogramming will cause Thrashing.**
- **Prior knowledge of scheduling algorithms (An algorithm that decides which next process will get hold of the CPU) is required. These are of two types:**

*** Preemptive Scheduling algorithm: if more than one process wants to enter into the critical section then it will be allowed and it can enter into the critical section without any interruption only if no other progress is in the critical section.**

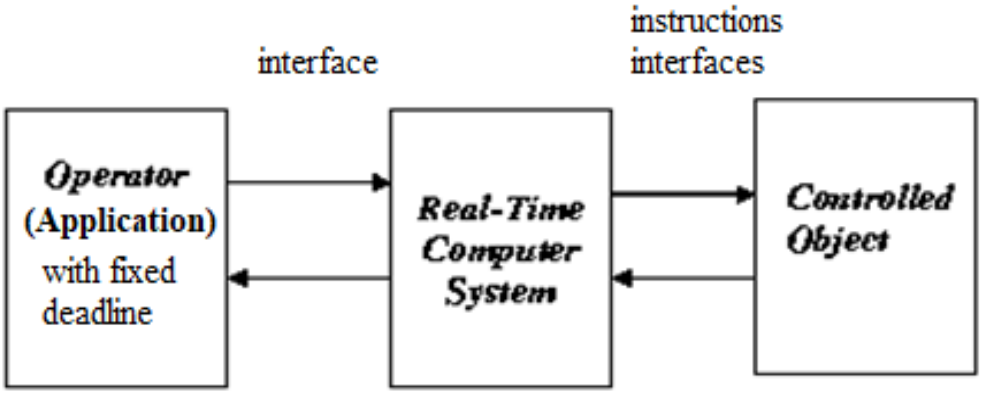
*** Non-Preemptive scheduling algorithm: If a process gets a critical section then it will not leave the critical section until or unless it works gets done.**

Evolution and types of OS:

View for Backend of OS

5. 1960s–1970s Real-Time Operating Systems (RTOS)

- **it is a multitasking operating system intended for applications with fixed deadlines (real-time computing).**
- **It is a special kind of operating system designed to handle tasks that need to be completed quickly and on time.**
- **Over the years, RTOS has improved with new technology and the need for reliable real-time performance.**
- **Such applications include some small embedded systems, automobile engine controllers, industrial robots, spacecraft, industrial control, and some large-scale computing systems.**



View for Backend of OS

5. 1960s–1970s Real-Time Operating Systems (RTOS)

The real time operating system can be classified into two categories:

1. **hard real time system and**
2. **soft real time system.**

A hard real-time system guarantees that critical tasks be completed on time. This goal requires that all delays in the system be bounded, from the retrieval of stored data to the time that it takes the operating system to finish any request made of it. Such time constraints dictate the facilities that are available in hard real time systems.

A soft real-time system is a less restrictive type of real-time system. Here, a critical real-time task gets priority over other tasks and retains that priority until it completes. Soft real time system can be mixed with other types of systems. Due to less restriction, they are risky to use for industrial control and robotics.

View for Backend of OS

5. 1960s–1970s Real-Time Operating Systems (RTOS)

Advantages

•**Maximum Consumption:** Maximum utilization of devices and systems. Thus more output from all the resources.

•**Task Shifting:** Time assigned for shifting tasks in these systems is very less. For example, in older systems, it takes about 10 microseconds. Shifting one task to another and in the latest systems, it takes 3 microseconds.

•**Focus On Application:** Focus on running applications and less importance to applications that are in the queue.

•**Real-Time Operating System In Embedded System:** Since the size of programs is small.

•**Error Free:** These types of systems are error-free.

•**Memory Allocation:** Memory allocation is best managed in these types of systems.

View for Backend of OS

5. 1960s–1970s Real-Time Operating Systems (RTOS)

Disadvantages

•**Limited Tasks:** Very few tasks run simultaneously, and their concentration is very less on few applications to avoid errors.

•**Use Heavy System Resources:** Sometimes the system resources are not so good and they are expensive as well.

•**Complex Algorithms** : The algorithms are very complex and difficult for the designer to write on.

•**Device Driver And Interrupt Signals:** It needs specific device drivers and interrupts signals to respond earliest to interrupts.

•**Thread Priority:** It is not good to set thread priority as these systems are very less prone to switching tasks.

•**Minimum Switching:** RTOS performs minimal task switching.

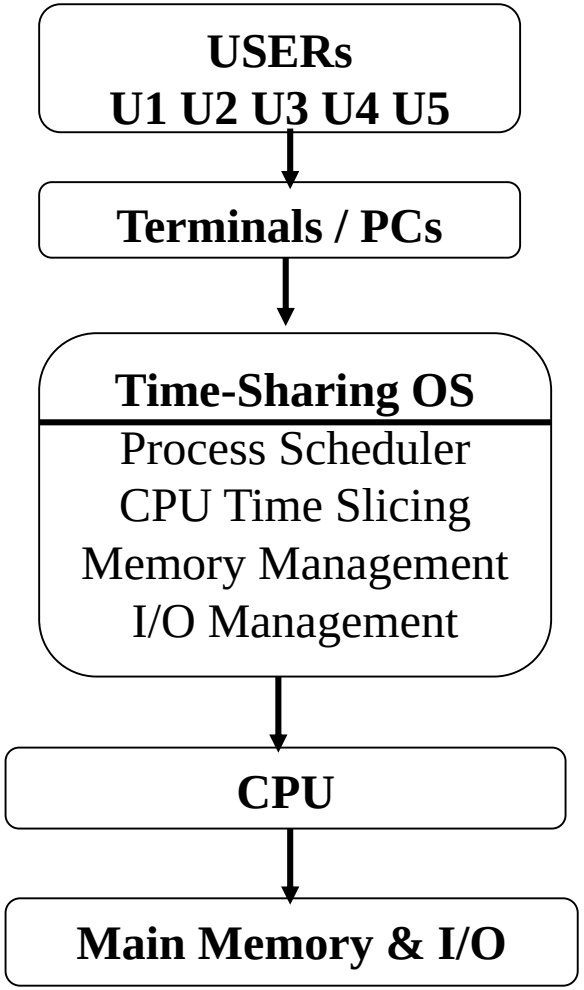
Evolution and types of OS:

View for Backend of OS

6. 1970s–1980s Time-Sharing/multitasking Operating Systems

Time sharing (or multitasking) OS is a logical extension of multiprogramming. It provides extra facilities such as:

- **Faster switching between multiple jobs to make processing faster.**
- **Allows multiple users to share computer system simultaneously.**
- **The users can interact with each job while it is running. These systems use a concept of virtual memory for effective utilization of memory space. Hence, in this OS, no jobs are discarded. Each one is executed using virtual memory concept. It uses CPU scheduling, memory management, disc management and security management. Examples: CTSS, MULTICS, CAL, UNIX etc.**



View for Backend of OS

6. 1970s–1980s Time-Sharing/multitasking Operating Systems

Advantages

1. Each task gets an equal opportunity.
2. Fewer chances of duplication of software.
3. CPU idle time can be reduced.

Disadvantages

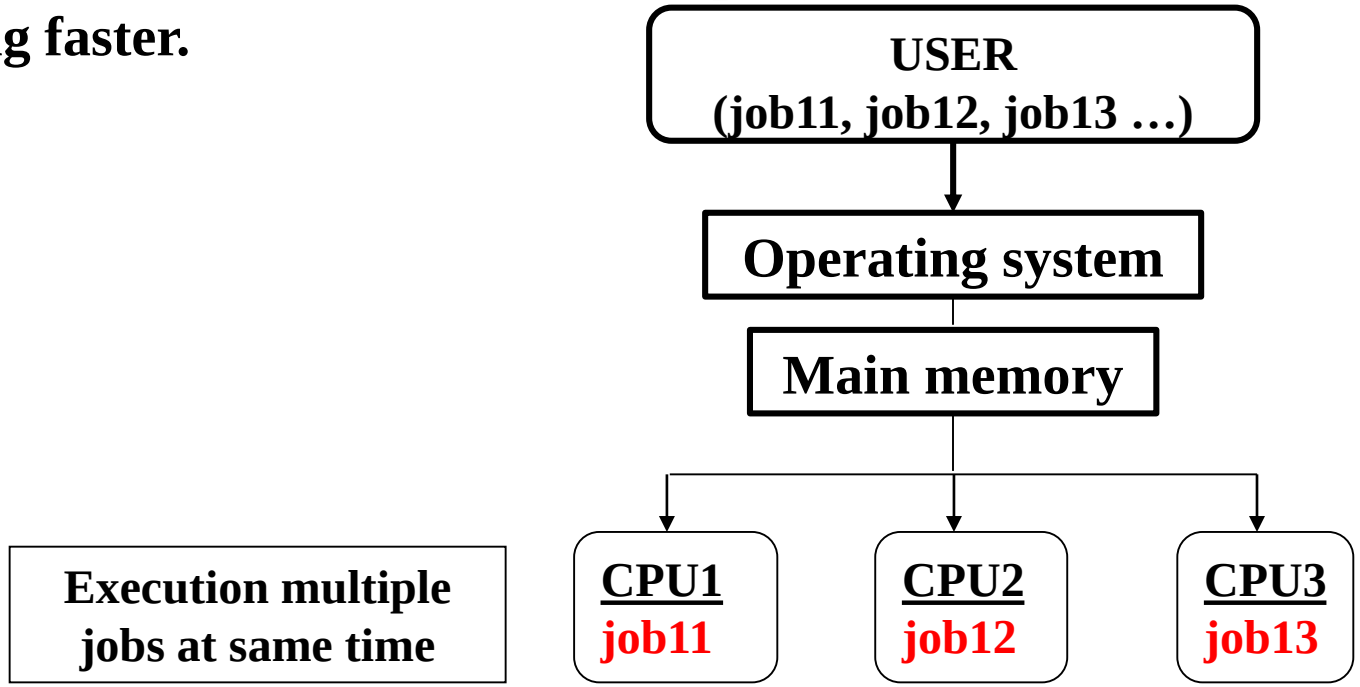
1. Reliability problem.
2. One must have to take of the security and integrity of user programs and data.
3. Data communication problem.

Evolution and types of OS:

View for Backend of OS

7. 1980s–1990s Multiprocessor Operating Systems

Multiprocessor operating systems are also known as parallel OS or tightly coupled OS. Such operating systems have more than one processor in close communication that sharing the computer bus, the clock and sometimes memory and peripheral devices. It executes multiple jobs at same time and makes the processing faster.



View for Backend of OS

7. 1980s–1990s Multiprocessor Operating Systems

Multiprocessor systems have three main advantages:

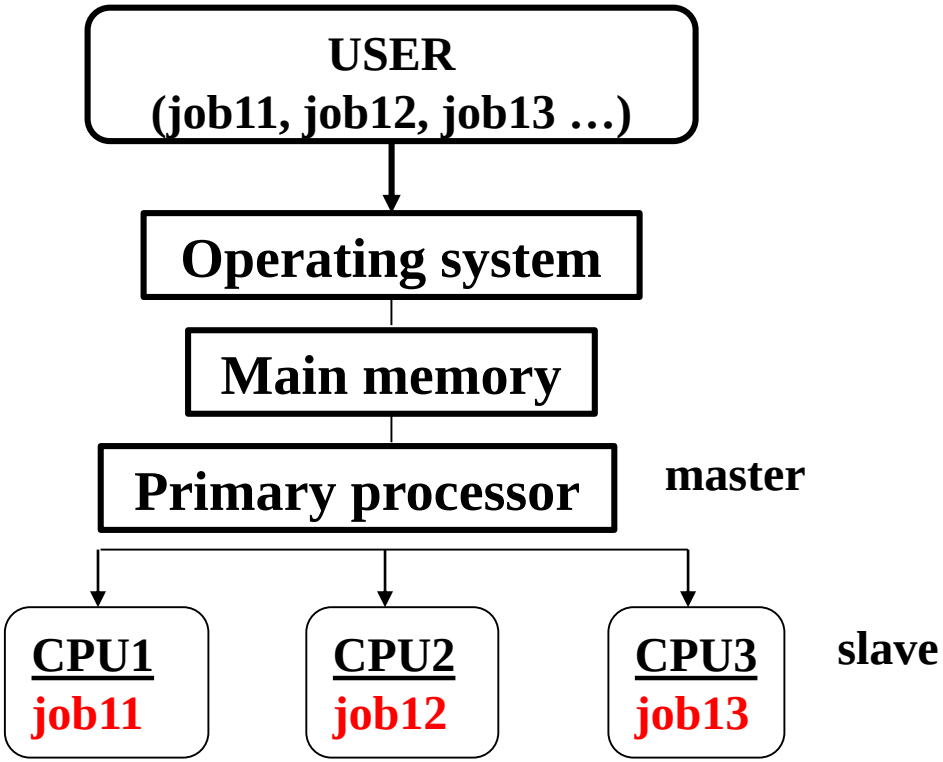
- **Increased throughput: By increasing the number of processors, the system performs more work in less time. The speed-up ratio with N processors is less than N.**
- **Economy of scale: Multiprocessor systems can save more money than multiple single-processor systems, because they can share peripherals, mass storage, and power supplies.**
- **Increased reliability: If one processor fails to done its task, then each of the remaining processors must pick up a share of the work of the failed processor. The failure of one processor will not halt the system, only slow it down.**

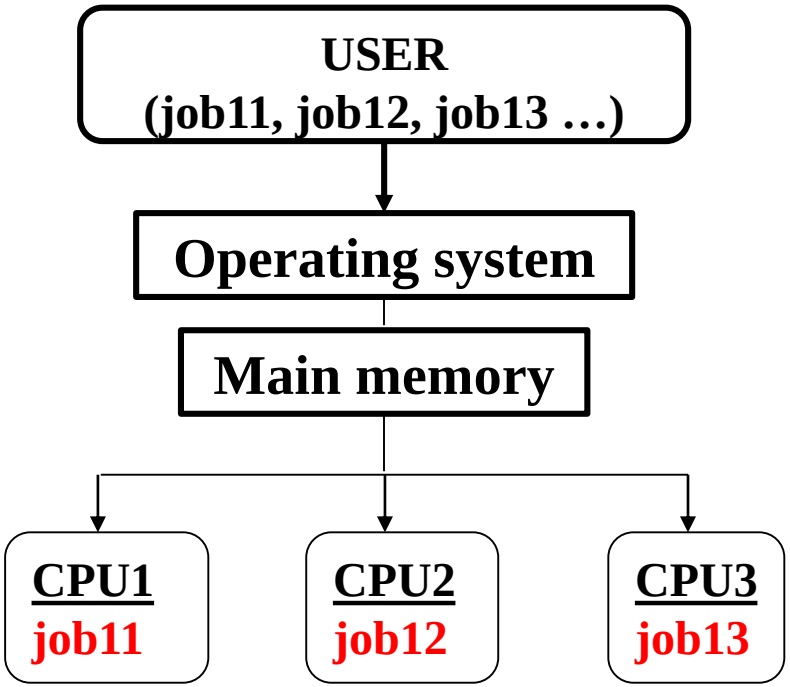
View for Backend of OS

7. 1980s–1990s Multiprocessor Operating Systems

The multiprocessor operating systems are classified into two categories:

1. **Symmetric multiprocessing system**
2. **Asymmetric multiprocessing system**





View for Backend of OS

7. 1980s–1990s Multiprocessor Operating Systems

The multiprocessor operating systems are classified into two categories:

1. Symmetric multiprocessing system

- **each processor runs an identical copy of the operating system, and these copies communicate with one another as needed.**

- **all processors are equal and execute the same instance of the operating system. Any processor can perform any task, including process scheduling and I/O handling.**
- **Processes are dynamically assigned to processors using CPU scheduling algorithms.**
- **All processors share the same physical memory and I/O subsystem.**
- **Also known as a Shared-Memory Multiprocessing System.**

View for Backend of OS

7. 1980s–1990s Multiprocessor Operating Systems

The multiprocessor operating systems are classified into two categories:

1. Symmetric multiprocessing system

Advantages

•Failure of one processor does not affect the functioning of other processors.
•It divides all the workload equally to the available processors.
•Makes use of available resources efficiently.

Disadvantages

•Symmetrical multiprocessing OS are more complex.
•They are more costlier.
•Synchronization between multiple processors is difficult.

View for Backend of OS

7. 1980s–1990s Multiprocessor Operating Systems

The multiprocessor operating systems are classified into two categories:

2. Asymmetric multiprocessing system

- **a processor is called master processor that controls other processors called slave processor. Thus, it establishes master-slave relationship. The master processor schedules the jobs and manages the memory for entire system.**
- **processors are not equal. One processor acts as the master, while the others act as slave processors.**
- **The master processor handles scheduling, memory management, and I/O operations.**
- **Slave processors execute tasks assigned by the master.**
- **The master maintains the ready queue and dispatches processes to slaves.**

View for Backend of OS

7. 1980s–1990s Multiprocessor Operating Systems

The multiprocessor operating systems are classified into two categories:

2. Asymmetric multiprocessing system

Advantages

•Asymmetrical multiprocessing operating system are cost-effective.
•They are easy to design and manage.
•They are more scalable.

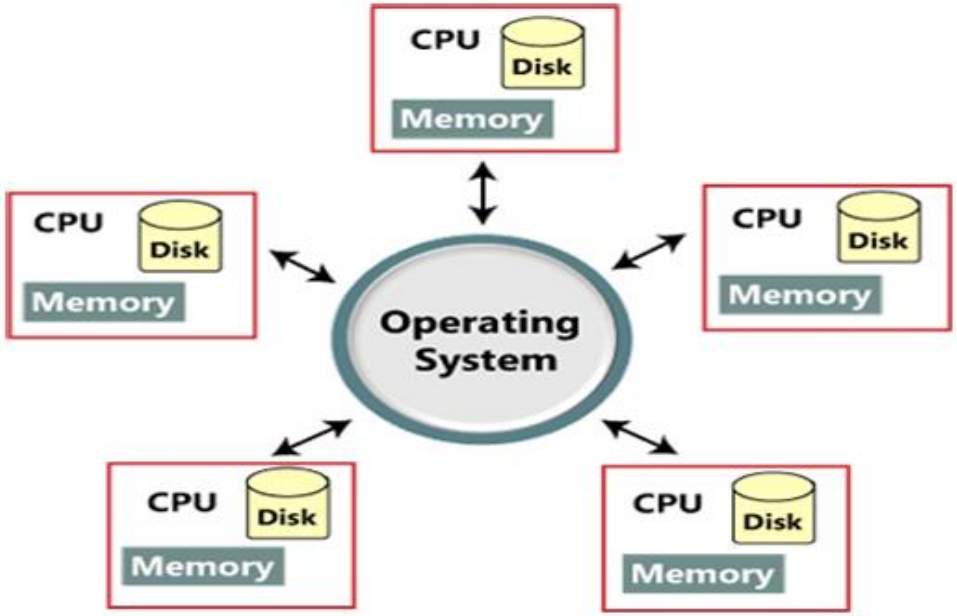
Disadvantages

•There can be uneven distribution of workload among the processors.
•The processors do not share same memory.
•Entire system goes down if one process fails.

View for Backend of OS

7. 2000 – present Distributed Operating Systems

- In distributed system, the different machines are connected in a network and each machine has its own processor and own local memory.
- In this system, the operating systems on all the machines work together to manage the collective network resource.



Evolution and types of OS:

View for Backend of OS

7. 2000 – present Distributed Operating Systems

- It can be classified into two categories:
  1. Client-Server systems
  2. Peer-to-Peer systems

  Advantages of distributed systems.
  - Resources Sharing
  - Computation speed up – load sharing
  - Reliability
  - Communications
  - Requires networking infrastructure.
  - Local area networks (LAN) or Wide area networks (WAN)

Evolution and types of OS:

View for Backend of OS

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Evolution and types of OS:

Types of OS in two views

- Backend of OS
- Frontend of OS

Evolution and types of OS:

View for Frontend of OS

1. Early Computers (1940s–1950s): No Interface

- Early computers had no operating system and no user interface.
- Programs were entered directly using machine language with switches or punched cards.
- Only highly trained operators could use these systems.

2. Batch Processing Interface (1950s–1960s)

- With the development of Batch Processing Operating System, users submitted jobs on punched cards.
- The computer processed jobs in batches without user interaction.
- The interface was indirect—users did not interact with the system while it was running.

Evolution and types of OS:

View for Frontend of OS

3. Command Line Interface (CLI) (1960s–1970s)

- Operating systems began allowing users to type commands.
- This was called the Command-Line Interface (CLI).
- Users entered text commands to control the system.
- Examples include systems like UNIX and MS-DOS.
- It required users to remember many commands, but it was powerful and efficient.

View for Frontend of OS

4. Graphical User Interface (GUI) (1980s–Present)

•A major improvement came with the Graphical User Interface (GUI).
•Users interact using windows, icons, menus, and a mouse instead of typing commands.
•The concept was first developed at Xerox PARC and later popularized by Apple Macintosh Operating System and Microsoft Windows.
•GUI made computers much easier for ordinary people to use.

5. Modern Interfaces (2000s–Present)

Modern operating systems support advanced interfaces such as:
•Touch interfaces (smartphones and tablets).
•Voice interfaces using assistants like Google Assistant and Siri.
•Gesture-based interfaces and AI-driven interactions.

Functions of Operating System:

Functions of Operating System:

Functions of Operating System:

1. Process Management
2. Main-Memory Management
3. File Management
4. I/O System Management
5. Secondary-Storage Management
6. Networking (Distributed Systems)
7. Protection System
8. Command-Interpreter System

Functions of Operating System:

1. Process Management

- **A process** in operating system can be defined as a program that is currently executing or running on a computer.
- It is an active instance that uses system resources such as CPU, memory, input, or output devices.
- **When a process is loaded into memory, it is divided into various sections that are used for different purposes:**

1. **Text or Code Segment**: The text segment contains the instructions of the compiled program, which is essentially the executable code of the program.
2. **Data Segment:** The data segment contains the global and static variables and is initialized by the program.
3. **Heap:** The heap is used for dynamic memory allocations that occur during the execution of the process.
4. **Stack:** The stack contains function call information, local variables, and return addresses. This means the stack is used for managing function calls and recursion.
5. **Memory-mapped Region:** The memory-mapped region contains shared libraries or files that are mapped into the address space of the process.

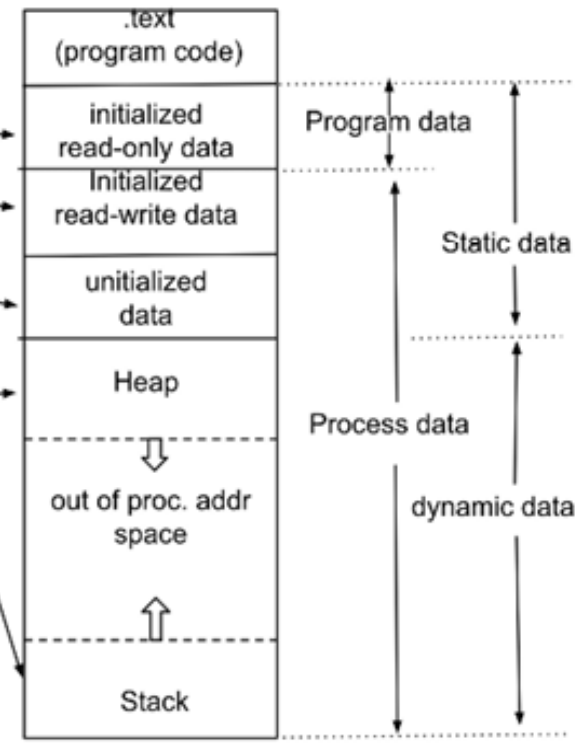
Functions of Operating System:

1. Process Management

- Each process has **its own process ID** and **address space.**

```
#include#include<stdio.h>
#include<stdlib.h>

const int x=5;
int y = 15;
int z;


int main(int argc, char *argv[])
{
   int *val;
   int i;

   val = (int *)malloc(sizeof(int)*10);

   for(i = 0; i < 10; i++)
      val[i] = i;

   return 0;
}
```

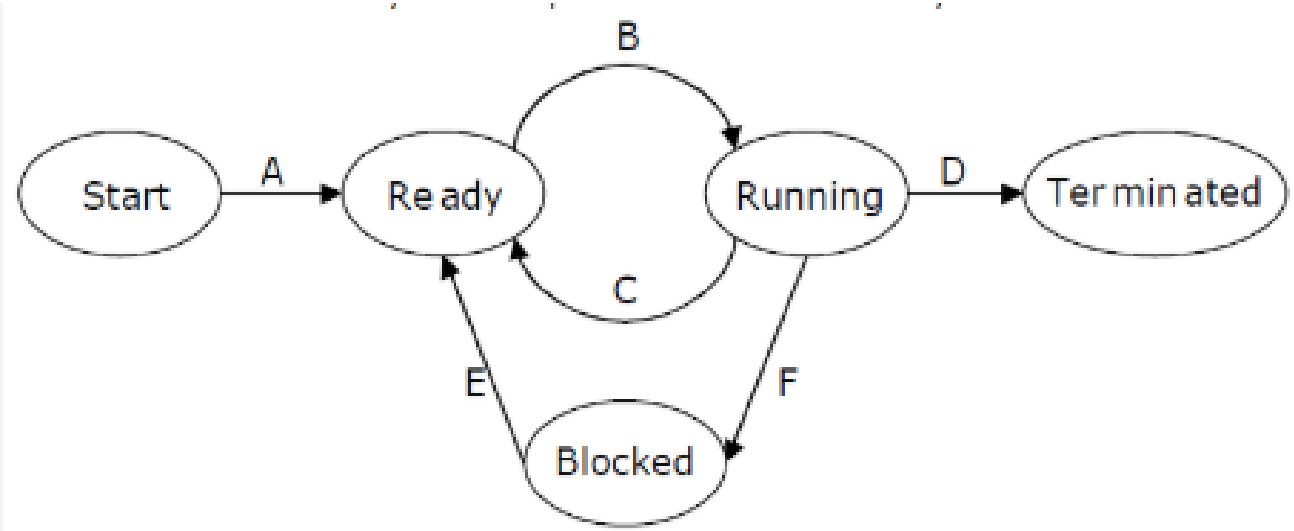
Address space



Functions of Operating System:

1. Process Management

- The processes within the operating system are managed by **process control blocks** (**PCBs**), which include **process states**, priority, and resources being used.
- **The processes** in the operating system can be in different states, including ready, running, and waiting.



waiting

Functions of Operating System:

1. Process Management

Here are the main states of the process in operating system:

- **New:** The state of the process where it is created, but is not ready for execution yet.
- **Ready:** It is the state of the process where it is loaded in memory and waiting for the CPU to execute the process.
- **Running:** The state where the process is being executed on the CPU.
- **Waiting (Blocked):** It is the state where the process cannot continue without an external event occurring, such as completing I/O.
- **Terminated (Exit):** It is the state in which the process has completed or failed execution.

The optional states that may be present in some systems:

- **Ready Suspended**: It is the state where the process is in secondary memory, but is ready to return to memory.
- **Waiting Suspended**: The state in which a process is both waiting for an event and also swapped out of memory.

Functions of Operating System:

1. Process Management

Process Life Cycle in Operating System

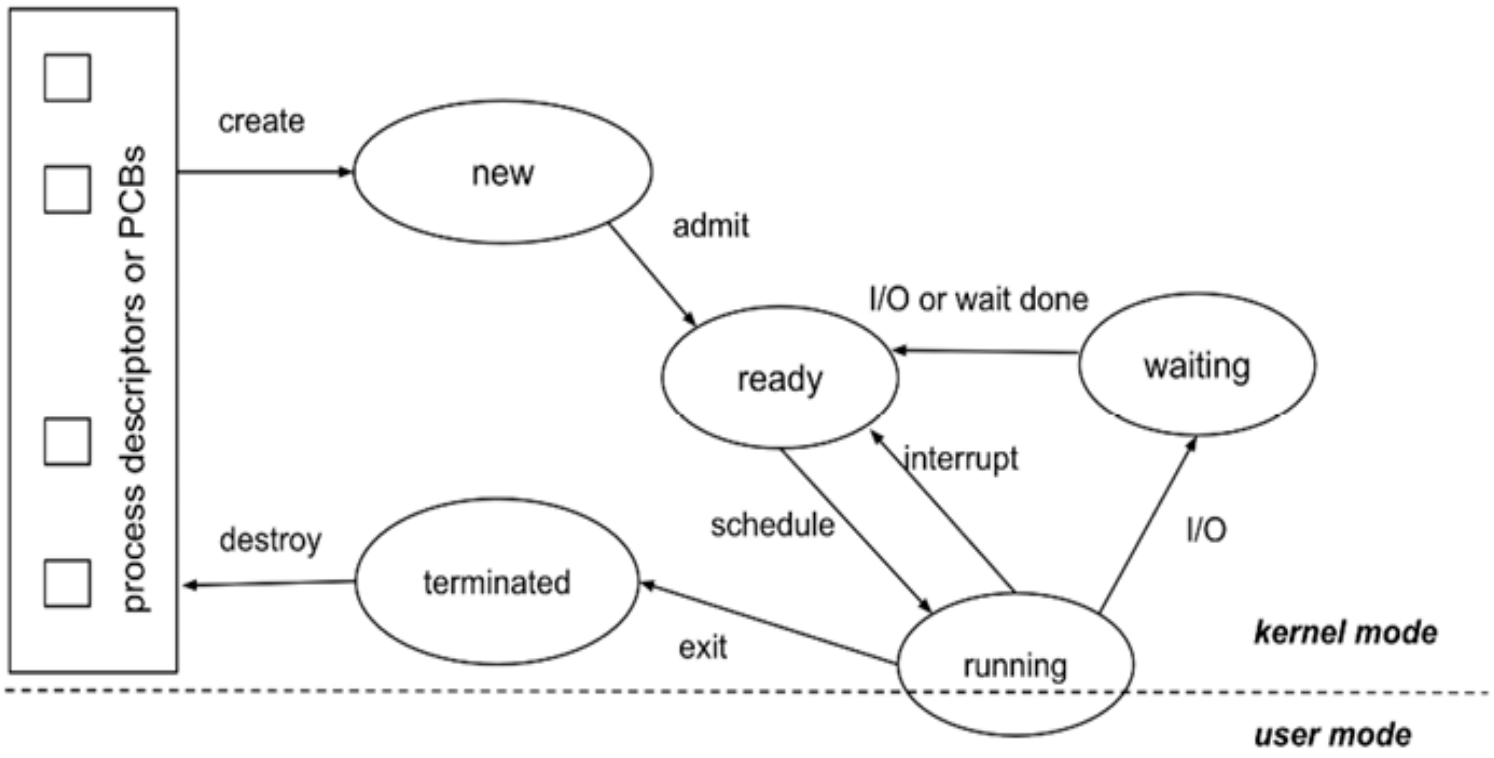
- The process life cycle is the cycle of the different stages a process goes through from when it is created until it finishes execution.

- When a program starts, it becomes a process and is in the New state.
- Then it moves to the ready state, waiting for the CPU. When the OS schedules the process, it goes into the running state to execute instructions.
- If the process needs to wait for something (input/output), it goes into the waiting (Blocked) state.
- After finishing the wait, it returns to the ready state.
- The process is in the terminated state when it finishes its task or has been terminated by the system.

Functions of Operating System:

1. Process Management

Process Life Cycle in Operating System



Functions of Operating System:

1. Process Management

Steps in the Process Life Cycle

1. **New**: The operating system creates and initializes the process.
2. **Ready:** The process is loaded into the main memory and is waiting for allocation of CPU time.
3. **Running:** The process is executing instructions on the CPU.
4. **Waiting or Blocked:** The process cannot continue past this point until an event occurs, like an I/O operation being completed.
5. **Ready (again):** After the waiting condition, the process is in the ready state and waiting for CPU time.
6. **Terminated or Exit:** The process has completed executing or is killed and is removed from the system.

Functions of Operating System:

1. Process Management

- **Processes** communicate with one another via inter-process communication (IPC) and are scheduled and managed by the OS for a given CPU time, or the time at which to run their instructions on the CPU.

Functions of Operating System:

1. Process Management

A software program is a set of instructions that are executed in the CPU to accomplish a specific task. A process needs certain resources, including CPU time, memory, files, and I/O devices, to accomplish its task.

The operating system is responsible for the following activities in connection with process management.

- ✦ **Process creation and deletion.**
- ✦ **process suspension and resumption.**
- ✦ **scheduling of processes (or threads aka sub-processes) for CPU time**
- ✦ **communication and coordination among several cooperating processes**
- ✦ **resolving contention among competing processes**
- ✦ **Provision of mechanisms for:**
  - **process synchronization**
  - **process communication**

Functions of Operating System:

2. Main-Memory Management

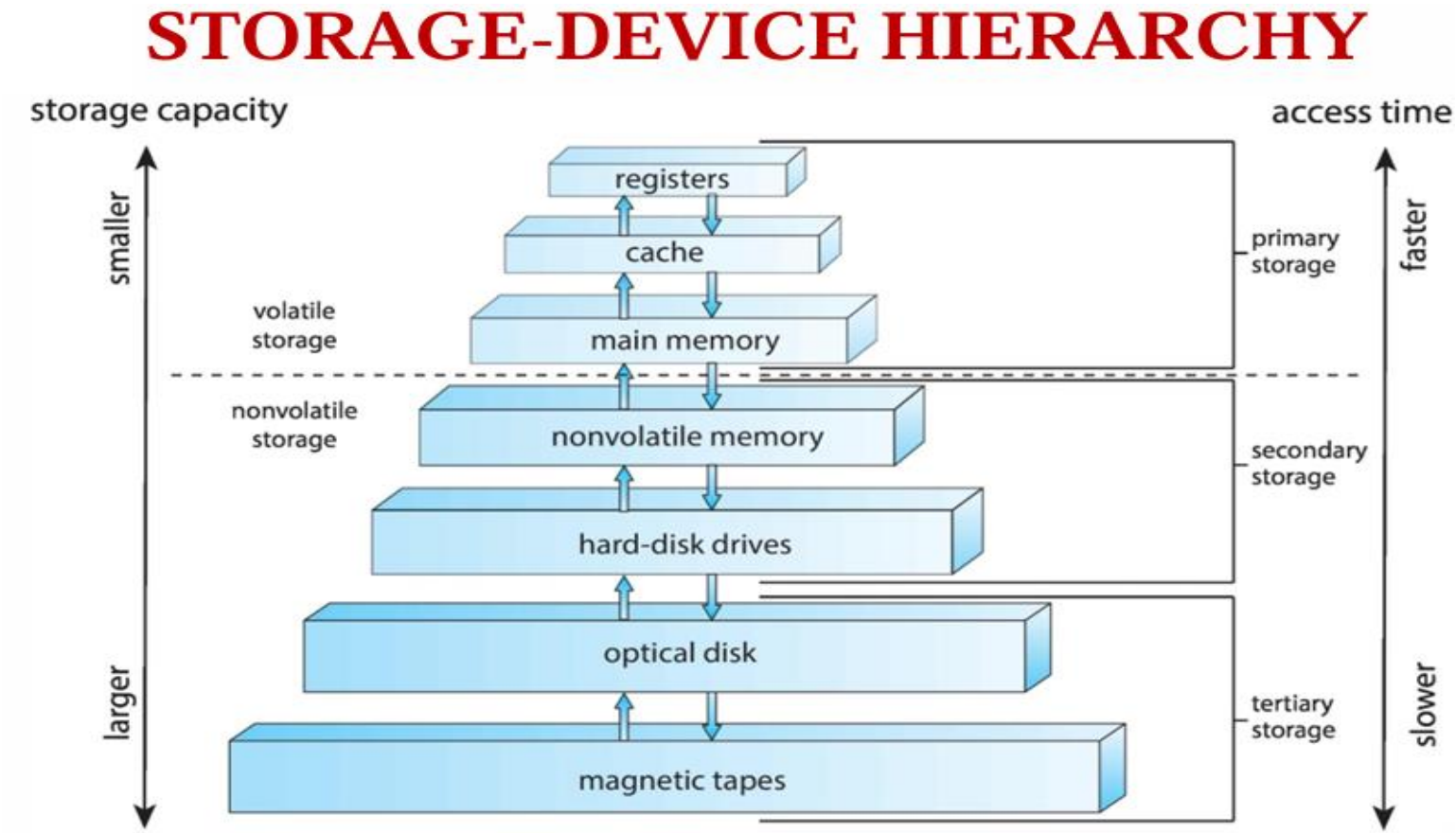
- **Memory is a large array of words or bytes, each byte having a fixed memory address.**
- **It is a repository of quickly accessible data shared by the CPU and I/O devices.**
- **Main memory is a volatile storage device. It loses its contents in the case of system failure.**
- **CPU fetches instructions from main memory addresses (during instruction-fetch cycle), accesses (read & write) data from there during the data-fetch cycle**

The operating system is responsible for the following activities in connections with memory management:

- **Keep track of which parts of memory are currently being used and by whom.**
- **Decide which processes to load when memory space becomes available.**
- **Allocate and de-allocate memory space as needed.**

Functions of Operating System:

2. Main-Memory Management



Functions of Operating System:

3. File Management

A file is a collection of related information defined by its creator. Commonly, files represent programs (both source and object forms) and data. The operating system is responsible for the following activities in connections with file management:

- ✦ **File creation and deletion.**
- ✦ **Directory creation and deletion.**
- ✦ **Support of primitives for manipulating files and directories.**
- ✦ **Mapping files onto secondary storage.**
- ✦ **File backup on stable (nonvolatile) storage media.**

Functions of Operating System:

4. I/O System Management

A computer can have several different I/O devices (keyboard, mouse, touchpad, styler pen, monitor, scanner, printer, external disk etc). These devices enter, store and/or retrieve data.

All application programs use I/O devices during their execution. These devices are made of different physical materials, have different physical characteristics, and thus require different handling techniques. Performance of a computer is dependent on proper management and control of the I/O devices and application programmers.

The drivers interact with the device-controllers and manage the devices. Operating systems provide users simpler interfaces to interact with the devices.

The I/O system consists of:

- ✦ **a memory management module to buffer, cache, and spool data transfer**
- ✦ **A general device-driver interface**
- ✦ **Drivers for specific hardware devices**

Functions of Operating System:

5. Secondary-Storage Management

Since main memory (primary storage) is volatile and too small to accommodate all data and programs permanently, the computer system must provide secondary storage to back up main memory. Most modern computer systems use disks as the principle on-line storage medium, for both programs and data.

The operating system is responsible for the following activities in connection with disk management:

- ✦ **Free space management**
- ✦ **Storage allocation**
- ✦ **Disk scheduling**

Functions of Operating System:

6. Networking (Distributed Systems)

- A distributed system is a collection processors that do not share memory or a clock. Each processor has its own local memory.
- The processors in the system are connected through a communication network.
- Communication takes place using a protocol.
- A distributed system provides user access to various system resources.
- Access to a shared resource allows:
  - ✦ Computation speed-up
  - ✦ Increased data availability
  - ✦ Enhanced reliability

Functions of Operating System:

7. Protection System

- **Protection refers to a mechanism for controlling access by programs, processes, or users to both system and user resources.**
- **The protection mechanism must:**
  - ✦ **distinguish between authorized and unauthorized usage.**
  - ✦ **specify the controls to be imposed.**
  - ✦ **provide a means of enforcement.**

Functions of Operating System:

8. Command-Interpreter System

- Many commands are given to the operating system by control statements which deal with:
  - ✦ process creation and management
  - ✦ I/O handling
  - ✦ secondary-storage management
  - ✦ main-memory management
  - ✦ file-system access
  - ✦ protection
  - ✦ networking
- The program that reads and interprets control statements is called variously:
  - ✦ command-line interpreter
  - ✦ shell (in UNIX)
- Its function is to get and execute the next command statement.